

Design and Implementation of an SDR-Based Wireless Channel Sounder for Indoor Multipath Characterization

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Abstract

Wireless communication systems operating in indoor environments are heavily impacted by multipath propagation, where transmitted signals reflect from walls, furniture, and other structures before arriving at the receiver. These reflections create delayed copies of the original waveform that can distort communication signals and degrade system performance. This project presents the design and implementation of a Software Defined Radio (SDR)-based wireless channel sounder capable of measuring and visualizing indoor multipath characteristics using Universal Software Radio Peripheral (USRP) hardware.

The implemented system utilizes two USRP B200 SDR platforms configured in a Single Input Single Output (SISO) architecture operating in the 2.45 GHz ISM band. A Pseudo-Random Binary Sequence (PN sequence) was transmitted as the probe waveform due to its favorable autocorrelation properties for channel estimation. Received IQ samples were processed offline using Python-based digital signal processing techniques, including cross-correlation, filtering, averaging, and frequency-domain analysis.

The project successfully demonstrated end-to-end wireless channel sounding and generated Channel Impulse Response (CIR) and Power Delay Profile (PDP) measurements across multiple environments and operating conditions. Experimental scenarios included line-of-sight (LOS), non-line-of-sight (NLOS), obstruction testing, distance sweeps, and bandwidth comparison analysis. Higher bandwidth configurations improved temporal resolution, reducing measured CIR peak width from approximately 8 μs at 250 kHz to approximately 0.6 μs at 5 MHz.

Results demonstrated measurable variations in delay spread, coherence bandwidth, peak-to-mean ratio, and frequency-selective fading behavior across LOS, NLOS, obstruction, and bandwidth-dependent propagation scenarios. The completed system validates the feasibility of a modular SDR-based channel sounder suitable for propagation experimentation, wireless systems analysis, and future RF research applications.

1. Introduction

1.1 Background

Modern wireless systems — including Wi-Fi, 5G, radar, tactical communications, and adaptive RF sensing platforms — operate in environments heavily influenced by multipath propagation effects. In indoor and dense urban environments, RF signals rarely travel directly from transmitter to receiver without interacting with surrounding structures, producing delayed reflections, scattering, frequency-selective fading, and signal distortion that can significantly alter wireless system performance. Understanding real-world wireless propagation behavior is therefore critical for the design, optimization, and validation of reliable wireless communication and sensing systems.

Wireless channel sounding is a method used to experimentally characterize propagation environments by transmitting a known waveform and analyzing how the channel alters the signal. By estimating the channel impulse response, researchers and engineers can visualize multipath structure and quantify propagation behavior.

This capstone project focused on developing a simplified but fully functional SDR-based wireless channel sounder capable of performing indoor propagation measurements and comparative RF analysis.

Prior research has demonstrated the effectiveness of PN-sequence-based channel sounding for wireless propagation analysis and delay spread characterization [1], while more recent SDR-based implementations have enabled flexible and low-cost experimental channel sounding platforms using USRP hardware and GNU Radio frameworks [3], [4]. This project builds upon those concepts through the implementation of a modular SDR-based indoor propagation measurement platform focused on comparative multipath analysis and frequency-selective characterization.

1.2 Problem Statement

Indoor RF propagation environments are complex and difficult to model analytically due to reflections and dynamic environmental conditions. Many professional channel sounding systems are expensive, proprietary, and difficult to modify for educational experimentation.

This project addresses the need for a flexible, low-cost, software-defined channel sounding platform capable of generating repeatable RF propagation measurements, measuring multipath delay structure, comparing LOS and NLOS environments, evaluating bandwidth-dependent resolution effects, and producing meaningful propagation visualizations.

Offline processing was intentionally selected to improve debugging flexibility, enable repeatable experimentation, and reduce synchronization complexity during development.

1.3 Project Objectives

The primary objectives of this project were:

1. Design and implement a functional SDR-based wireless channel sounding system
2. Transmit and receive a PN sequence using USRP B200 SDR hardware
3. Capture and process IQ data for channel estimation
4. Generate Channel Impulse Response (CIR) measurements
5. Produce Power Delay Profile (PDP) visualizations
6. Compare propagation behavior across multiple environments
7. Evaluate bandwidth-dependent delay resolution effects
8. Implement advanced analysis extensions, including FFT-based channelization

1.4 Scope

The project focused on indoor wireless channel characterization using a SISO architecture at 2.45 GHz. The system was intentionally scoped to prioritize the successful implementation of a complete end-to-end signal processing pipeline within the project timeline.

Included capabilities:

- SDR-based transmit and receive operation
- PN-sequence waveform generation
- Offline signal processing
- CIR and PDP generation
- LOS/NLOS comparison
- Distance-based testing
- Bandwidth comparison analysis
- FFT-based frequency-selective analysis

Deferred capabilities:

- MIMO channel sounding
- Real-time synchronization improvements
- Doppler analysis
- Multi-band operation
- Outdoor large-scale propagation modeling

2. System Theory and Technical Background

2.1 Multipath Propagation

In indoor RF environments, transmitted signals propagate along multiple paths before arriving at the receiver. These propagation paths may include direct LOS components, wall reflections, metallic reflections, scattering from indoor objects, and partial diffraction around obstructions.

Each propagation path introduces a different delay and attenuation. The superposition of these signals forms the received waveform.

This discrete-time channel model aligns directly with standard digital communications signal-processing frameworks used in SDR-based systems [11].

The received signal may be modeled as:

$$r(t) = \sum_{i=1}^N a_i s(t - \tau_i)$$

where:

- (a_i) represents path attenuation
- (τ_i) represents propagation delay
- $(s(t))$ is the transmitted waveform

While the continuous-time representation provides conceptual insight into multipath propagation behavior, practical SDR implementations operate on discretely sampled IQ data. The wireless channel can therefore also be modeled using a discrete-time convolution framework:

$$r[n] = \sum_{k=0}^{L-1} h[k] s[n - k] + w[n]$$

where:

- $h[k]$ represents the discrete channel impulse response
- $w[n]$ represents additive noise

This discrete-time channel model aligns directly with the implemented SDR processing architecture and supports offline channel estimation using captured IQ samples.

2.2 Channel Impulse Response

The Channel Impulse Response (CIR) describes how the wireless channel modifies a transmitted signal over time. Peaks in the CIR correspond to dominant propagation paths.

The CIR provides insight into delay spread, multipath structure, relative path strength, and environmental reflection behavior. Channel impulse responses were estimated using cross-correlation between the transmitted PN sequence and the received IQ samples.

$$R_{sr}[m] = \sum_n s[n] r^*[n + m]$$

where:

- $R_{sr}[m]$ is the cross-correlation between transmitted and received signals
- $s[n]$ is the transmitted PN sequence
- $r[n]$ is the received signal
- m is the sample delay offset
- $(\cdot)^*$ denotes complex conjugation

Cross-correlation-based PN sequence detection operates similarly to a matched-filter process, where correlation peaks correspond to estimated propagation delays and dominant signal paths[17].

$$\hat{\tau} = \arg \max_m |R_{sr}[m]|$$

where $\hat{\tau}$ represents the estimated propagation delay corresponding to the dominant correlation peak.

Propagation delay estimates were determined from the location of dominant cross-correlation peaks within the estimated CIR. Correlation floor behavior was influenced by noise, environmental multipath structure, and finite observation length effects. Correlation-based delay estimation can also be interpreted as a parameter estimation problem within classical statistical signal processing frameworks [17].

2.3 Power Delay Profile

The Power Delay Profile (PDP) is derived from the squared magnitude of the CIR and represents received signal power as a function of propagation delay.

$$P(\tau) = |h(\tau)|^2$$

The discrete-time representation may be expressed as:

$$P[k] = |h[k]|^2$$

where $P(\tau)$ and $P[k]$ represent received signal power as a function of propagation delay, and $h(\tau)$ and $h[k]$ represent the continuous and discrete channel impulse responses,

respectively. In the implemented SDR processing workflow, these delay-domain quantities were derived from relative cross-correlation sample alignment rather than calibrated absolute propagation timing references. Accordingly, the extracted RMS spread metrics are interpreted primarily as comparative correlation-domain dispersion measurements.

The PDP provides a useful representation of multipath energy distribution and supports evaluation of Correlation-Domain RMS Spread, channel dispersion, coherence bandwidth estimates, and overall multipath richness.

In this work, Correlation-Domain RMS Spread refers to RMS dispersion metrics derived from relative cross-correlation delay structure rather than calibrated absolute physical propagation delay measurements.

2.4 PN Sequence Channel Sounding

PN-sequence-based channel sounding is widely used because the autocorrelation properties of maximum-length sequences approximate an impulse response, enabling practical channel impulse response estimation through cross-correlation processing [16].

A Pseudo-Random Binary Sequence (PN sequence) was selected as the sounding waveform because of its favorable autocorrelation properties.

The PN sequence provided several advantages for channel sounding, including sharp correlation peaks, improved noise immunity, efficient impulse-response estimation, and repeatable waveform generation.

Cross-correlation between the transmitted and received sequences enables estimation of the wireless channel response.

$$\Delta\tau \approx \frac{1}{B}$$

where:

- $\Delta\tau$ = delay resolution
- B = signal bandwidth

This relationship demonstrates that increasing signal bandwidth improves temporal resolution and enables finer separation of closely spaced multipath components.

$$B_c \approx \frac{1}{5\tau_{\text{RMS}}}$$

A commonly used approximation is:

$$B_c \propto \frac{1}{\tau_{\text{RMS}}}$$

where B_c represents coherence bandwidth and τ_{RMS} represents the RMS delay spread.

Increased delay spread generally reduces coherence bandwidth, indicating stronger frequency-selective fading behavior across the wireless channel.

2.5 Bandwidth and Frequency-Selective Analysis

In addition to delay-domain analysis, wireless channels may also exhibit frequency-selective fading behavior caused by multipath propagation and channel dispersion. Delay spread characterization can be performed by transmitting pseudo-random sequences through the wireless channel and estimating the channel impulse response through cross-correlation processing. Variations in delay spread and CIR structure observed during obstruction testing are consistent with expected multipath behavior in dynamically changing propagation environments.

Wideband channel sounding enables characterization of wireless channels across both time and frequency domains, supporting analysis of delay spread, coherence bandwidth, and frequency-selective fading behavior in dynamically varying propagation environments [2]. Wireless channels vary across time, frequency, and space due to multipath propagation effects and environmental geometry.

In frequency-selective channels, delayed multipath components produce constructive and destructive interference across frequency, causing portions of the spectrum to experience different attenuation levels. FFT-based sub-band analysis, therefore, provides additional insight beyond time-domain CIR estimation by revealing how propagation effects vary across frequency. This behavior becomes increasingly important in wideband wireless systems where channel response cannot be assumed flat across the occupied bandwidth.

2.6 SDR-Based Channel Sounding Platforms

SDR-based channel sounders provide practical propagation measurement capability while significantly reducing the cost and complexity associated with traditional proprietary RF measurement systems. USRP platforms combined with GNU Radio have become widely used for experimental wireless channel sounding due to their flexibility, reconfigurability, and rapid prototyping capability. These advancements have enabled accessible and modular wireless channel sounding architectures using commercially available SDR hardware and open-source signal processing frameworks.

Low-cost SDR platforms provide accessible channel sounding capability for wireless propagation experimentation while introducing practical synchronization and timing challenges that must be mitigated in software.

Building upon these theoretical concepts, the following sections describe the implemented SDR hardware architecture and signal-processing workflow used throughout the experimental campaign. SDR-based indoor propagation measurement systems operating in the 2.4 GHz ISM band have also demonstrated the effectiveness of low-cost SDR platforms for practical channel sounding experimentation [6].

3. System Architecture and Design

3.1 Overall System Architecture

The implemented system consisted of:

- Transmit node
- Wireless propagation channel
- Receive node
- Offline signal processing pipeline
- Analysis and visualization framework

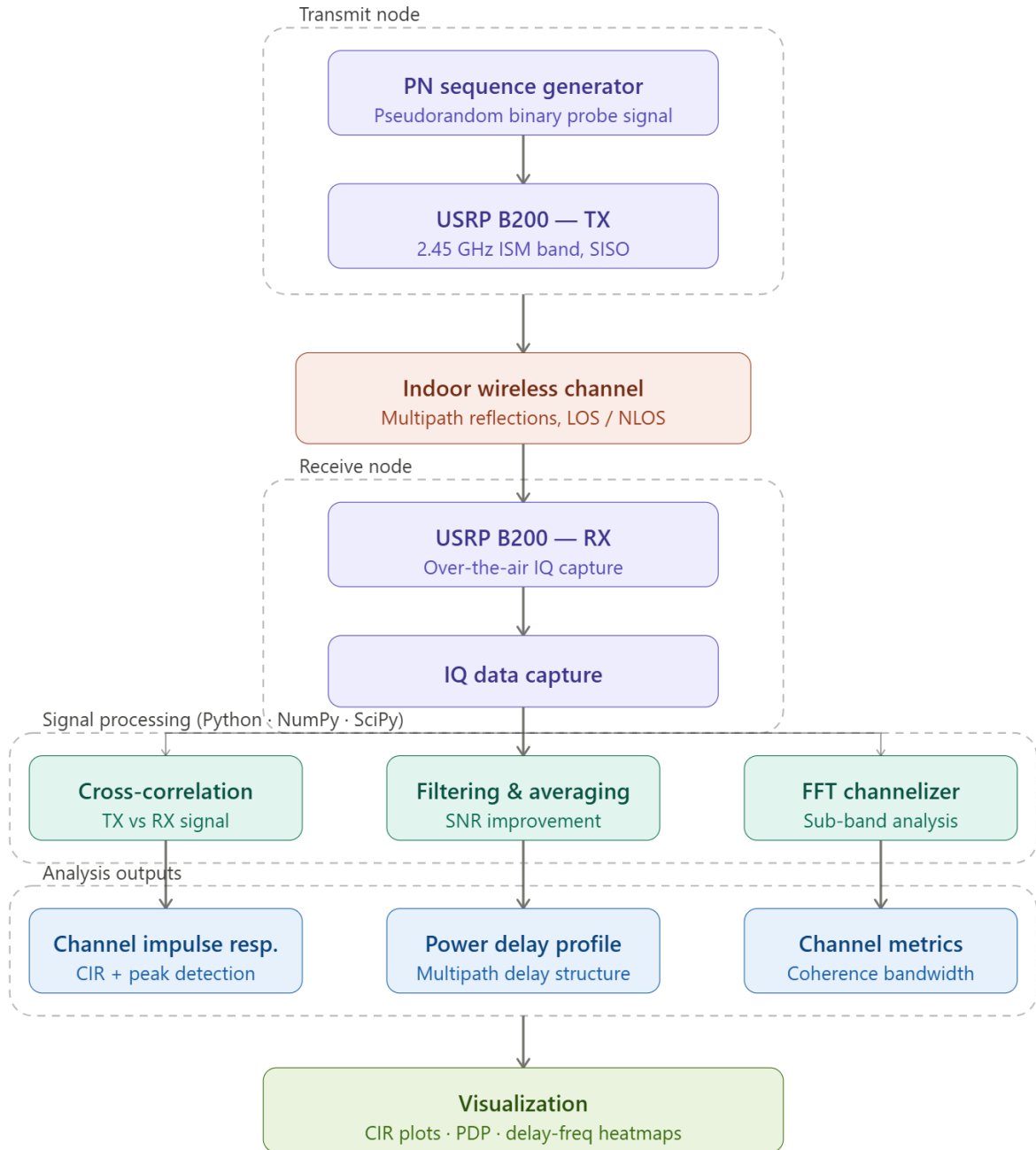


Figure 1. End-to-end SDR-based wireless channel sounding architecture showing PN-sequence transmission, USRP-based RF capture, offline signal processing, and propagation analysis workflow.

3.2 Hardware Configuration

SDR Platforms

The system utilized two Ettus Research USRP B200 SDR devices:

- One configured as a transmitter

- One configured as receiver

Key hardware characteristics:

- Frequency range: 70 MHz – 6 GHz
- USB 3.0 interface
- Full duplex capability
- Compact SDR architecture

Antennas

Two 2.4 GHz omnidirectional monopole antennas were used for over-the-air measurements.

Operating Parameters

Parameter	Value
Frequency	2.45 GHz
Architecture	SISO
Environment	Indoor
Waveform	PN Sequence
Processing Language	Python

Table 1. Primary SDR operating parameters used throughout experimental testing.

3.3 Software Environment

The signal processing and analysis pipeline was implemented in Python using:

- NumPy
- SciPy
- Matplotlib

Optional GNU Radio integration was evaluated during early development stages.

3.4 Signal Processing Pipeline

The implemented signal processing chain included:

1. PN sequence generation
2. SDR transmission
3. IQ capture
4. Cross-correlation
5. CIR extraction
6. PDP generation
7. Filtering and averaging

8. Comparative analysis
9. Visualization

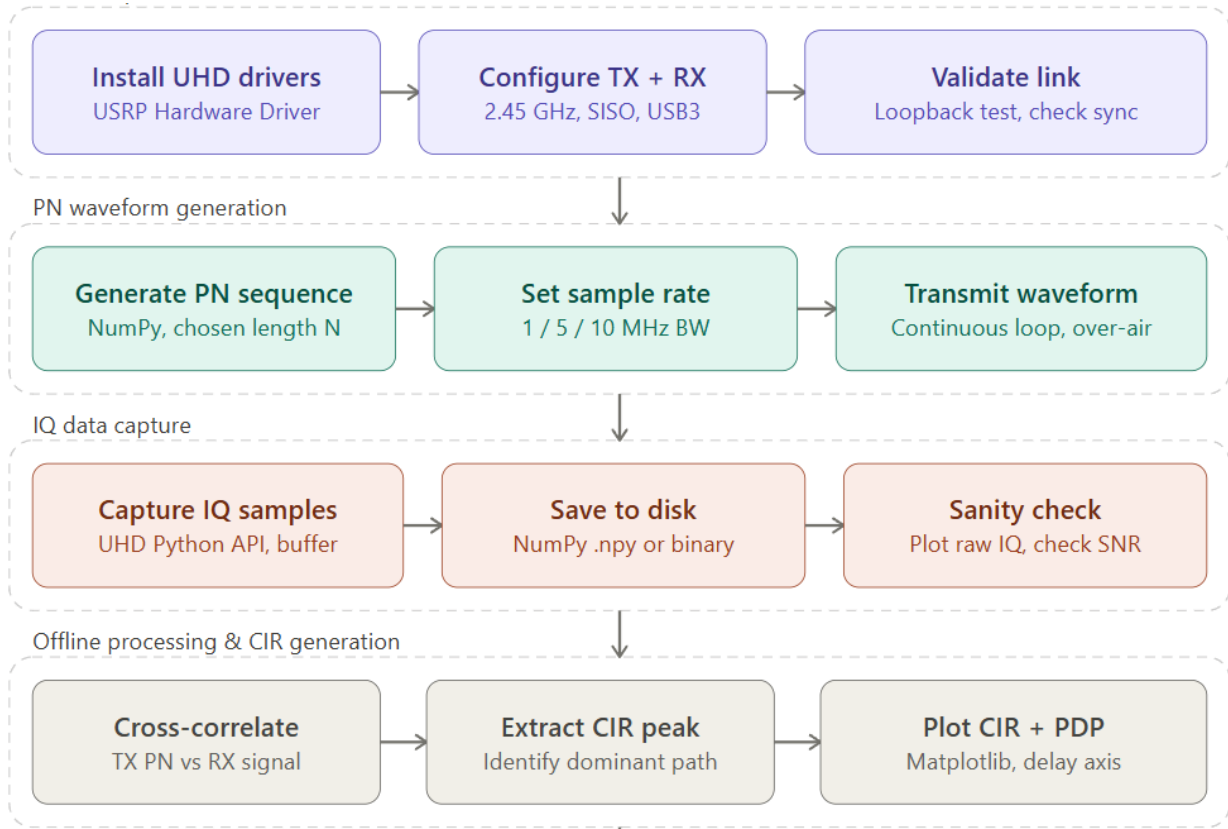


Figure 2. Signal-processing workflow used for SDR-based channel sounding, including PN waveform generation, IQ sample capture, offline cross-correlation, and CIR/PDP extraction.

Following system development, a series of controlled propagation experiments were conducted to evaluate channel behavior across multiple environmental conditions.

4. Implementation Methodology

4.1 PN Sequence Generation

A pseudo-random binary sequence was generated and used as the sounding waveform. The PN waveform was selected to maximize correlation detectability and support robust impulse response estimation. The generated sequence was transmitted repeatedly to allow coherent averaging and improved measurement stability. The implemented PN waveform utilized a degree-10 maximum-length pseudo-random binary sequence (m-sequence) of length $N = 1023$ samples generated using NumPy-based binary sequence generation methods.

4.2 SDR Transmission and Reception

The transmitter USRP B200 continuously transmitted the PN waveform at 2.45 GHz. The receiver captured raw complex IQ samples for offline analysis.

Initial validation included:

- USB connectivity verification
- UHD driver configuration
- Gain tuning
- Over-the-air signal verification

4.3 IQ Data Capture

Received IQ data was stored for offline processing and repeatable analysis. This allowed multiple processing approaches to be tested without requiring repeated RF captures.

4.4 Cross-Correlation Processing

Cross-correlation between the transmitted PN sequence and received IQ samples was performed to estimate the CIR. The dominant correlation peak was used as the primary timing reference for CIR extraction and comparative metric analysis. The cross-correlation output produced identifiable peaks corresponding to dominant propagation paths.

4.5 Filtering and Averaging

Additional processing enhancements included noise filtering, smoothing, averaging, and automated peak extraction to improve CIR clarity and reduce environmental noise sensitivity.

4.6 FFT-Based Channelization

An FFT-based channelizer was implemented as a stretch objective to evaluate frequency-selective fading behavior across the occupied signal bandwidth. This processing approach enabled sub-band channel estimation, frequency-response analysis, delay-versus-frequency heatmap visualization, and coherence bandwidth estimation across multiple propagation scenarios.

5. Experimental Setup

5.1 Experimental Matrix

The following experiments were conducted to evaluate indoor wireless propagation behavior across multiple environmental and signal-processing conditions.

Experiment	Variables Evaluated	Primary Metrics
LOS vs NLOS Comparison	Obstruction presence	CIR, PDP, delay spread
Distance Sweep	TX-RX separation distance	RMS delay spread, peak ratio
Object Obstruction Testing	Material and obstruction type	Peak-to-mean ratio, CIR distortion
Bandwidth Comparison	SDR bandwidth configuration	Temporal resolution, multipath separation
FFT/Sub-Band Analysis	Frequency-selective behavior	Sub-band response, coherence bandwidth

5.2 Test Environments

Experiments were conducted across multiple indoor propagation scenarios.

Baseline LOS Environment

The transmitter and receiver were positioned with minimal obstruction between antennas.

NLOS and Obstruction Testing

Additional experiments introduced obstructions, including:

- Cardboard
- Aluminum
- Environmental structures
- Indoor obstacles

Distance Sweep Testing

Distance-based testing evaluated propagation behavior at approximately 6 ft, 10 ft, and 15 ft transmitter-receiver separation distances.

5.3 Testing Procedure

For each environment:

1. Configure SDR parameters
2. Transmit PN waveform
3. Capture IQ samples
4. Perform offline processing
5. Generate CIR and PDP
6. Extract channel metrics
7. Compare results across scenarios

5.4 Validation Strategy

Validation included:

- Repeatability testing
- Multiple environment comparisons
- Correlation peak verification
- Consistent CIR extraction

The resulting experimental data were then processed and analyzed to evaluate multipath behavior, delay spread characteristics, and frequency-selective effects. Repeated measurements within the same static environment generally produced consistent CIR structures and Correlation-Domain RMS Spread variation within approximately 5–10%, depending on environmental stability and receiver positioning.

6. Results and Analysis

6.1 Initial Correlation and CIR Validation

The system successfully generated identifiable cross-correlation peaks and measurable channel impulse responses.

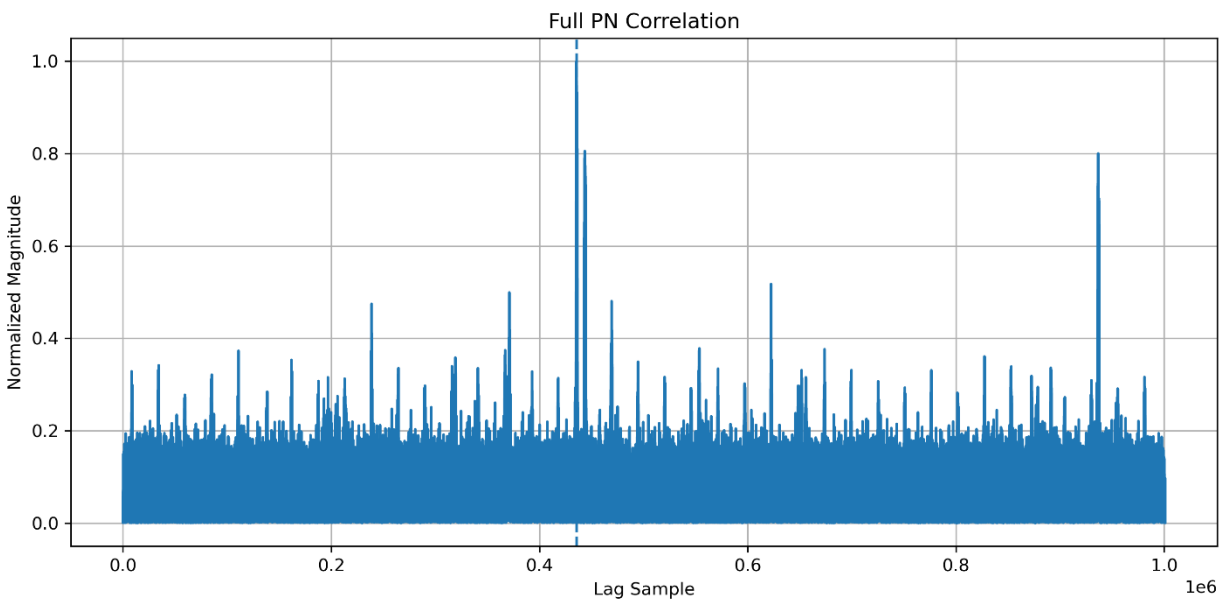


Figure 3. Full PN-sequence cross-correlation output showing successful synchronization and dominant correlation peak detection during channel sounding operation.

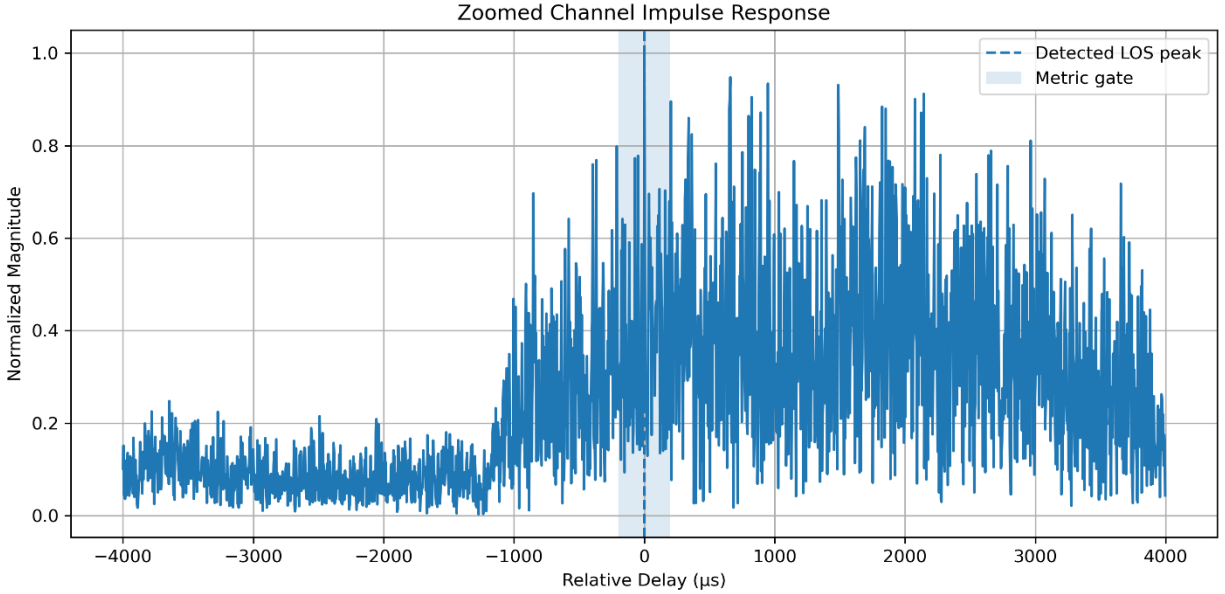


Figure 4. Zoomed channel impulse response (CIR) illustrating dominant propagation paths and secondary multipath reflections within the indoor environment. The displayed delay axis represents relative correlation-domain sample delay derived from SDR sample indexing and cross-correlation alignment rather than calibrated physical propagation delay.

The presence of dominant peaks verified successful synchronization and end-to-end channel sounding operation. The relatively elevated correlation floor and secondary peak structure observed in several measurements were influenced by environmental multipath reflections, SDR noise, and finite observation-window effects. Averaging and smoothing operations were therefore applied during later comparative visualizations to improve readability and emphasize dominant propagation behavior.

6.2 Comparative Indoor Propagation Behavior

Indoor propagation experiments demonstrated that environmental geometry and obstruction placement significantly altered multipath structure and channel behavior.

LOS environments generally produced stronger dominant propagation paths with lower delay spread and reduced multipath dispersion. In contrast, NLOS and obstructed environments produced broader CIR structures, elevated noise floors, and stronger secondary reflection behavior caused by indirect propagation paths.

The displayed delay axes represent relative correlation delay derived from SDR sample indexing and cross-correlation alignment rather than calibrated absolute propagation delay measurements. As a result, the reported delay-domain values primarily support comparative multipath analysis across experimental scenarios rather than absolute physical path-delay estimation.

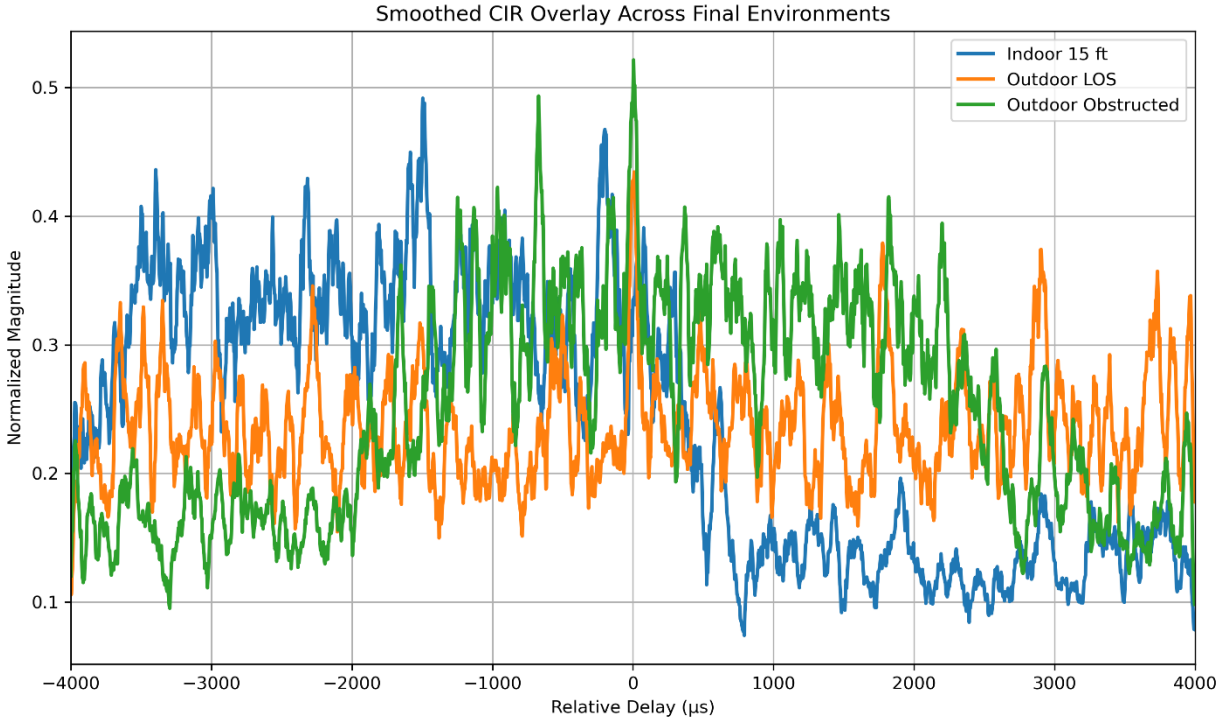


Figure 5. Smoothed CIR overlays comparing indoor LOS, outdoor LOS, and outdoor obstructed propagation environments. Obstructed environments produced broader multipath structure, elevated delay spread, and stronger secondary reflections relative to simpler LOS conditions.

Light moving-average smoothing was applied only for visualization clarity and did not alter the underlying extracted channel metrics.

The overlay comparisons illustrate increased delay dispersion and richer reflection behavior in obstructed environments relative to baseline LOS conditions.

6.3 Spatial Sensitivity and Distance Variation

Indoor distance variation experiments demonstrated the spatial sensitivity of multipath propagation, where relatively small receiver position changes produced measurable differences in delay spread and dominant path structure.

Increasing transmitter-receiver separation measurably affected peak signal amplitude, Correlation-Domain RMS Spread, peak-to-mean ratio, and background correlation behavior across the tested indoor environments. Correlation-Domain RMS Spread varied from approximately 114 μs at 6 ft, decreased to approximately 105 μs at 10 ft, and increased again to approximately 119 μs at 15 ft, demonstrating non-monotonic propagation behavior within the indoor environment.

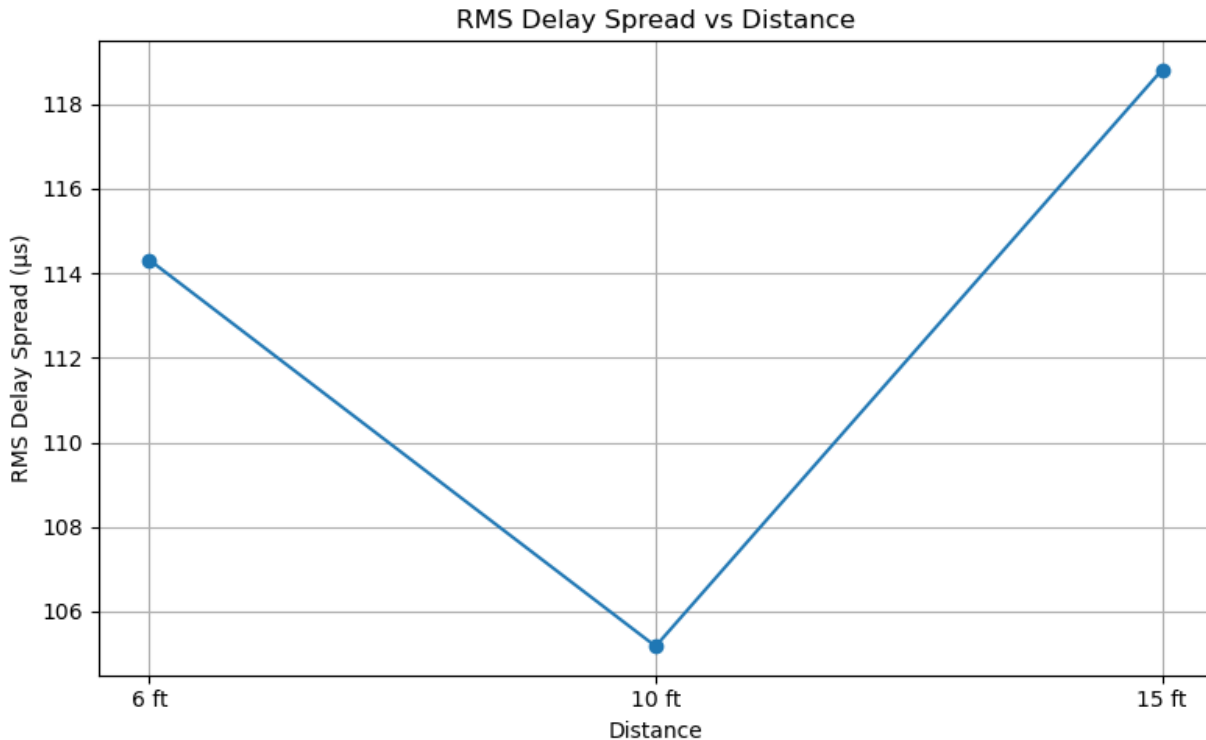


Figure 6. Correlation-Domain RMS Spread variation across indoor distance measurements. Small receiver position changes produced measurable differences in multipath dispersion and dominant-path behavior.

The indoor measurement environment consisted of an open-concept residential condo layout containing interconnected living room, kitchen, hallway, wall, floor, ceiling, furniture, and appliance reflection paths. Because the transmitter and receiver positions traversed a multipath-rich indoor geometry with numerous reflective surfaces at varying distances and angles, relatively small receiver position changes likely altered constructive and destructive interference relationships between dominant reflected signal components.

The non-monotonic reduction in both Correlation-Domain RMS Spread and peak-to-mean ratio observed at the 10 ft receiver position likely resulted from localized indoor multipath geometry effects, such as partial destructive cancellation of secondary reflections or temporary alignment with a local reflection null. Because both metrics exhibited similar behavior at the same receiver position, the observed reduction likely resulted from a consistent environmental geometry effect rather than random measurement variability. Although the mean correlation floor increased at the 10 ft position, the simultaneous reduction in dominant-path strength and Correlation-Domain RMS Spread suggests that localized interference effects redistributed energy across weaker reflected components rather than eliminating multipath behavior entirely.

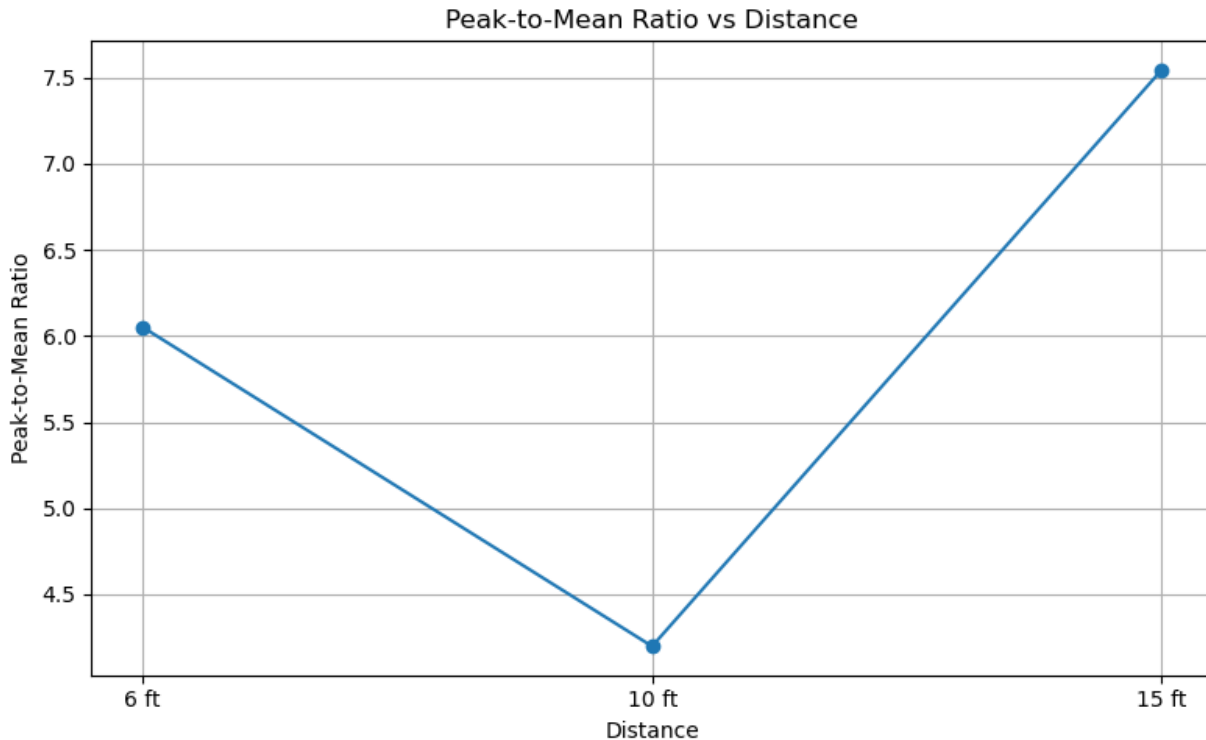


Figure 7. Peak-to-mean ratio variation across indoor distance measurements. Constructive and destructive interference effects caused non-monotonic changes in dominant path strength across receiver positions.

6.4 Obstruction and Material Effects

Different obstruction materials produced distinct propagation effects. Metallic obstructions generally produced higher attenuation, stronger reflection artifacts, and increased multipath distortion relative to non-metallic obstructions. For example, the aluminum obstruction produced a peak-to-mean ratio of approximately 25, compared to approximately 16 for cardboard and wood obstructions, indicating substantially stronger dominant reflection behavior and altered multipath structure.

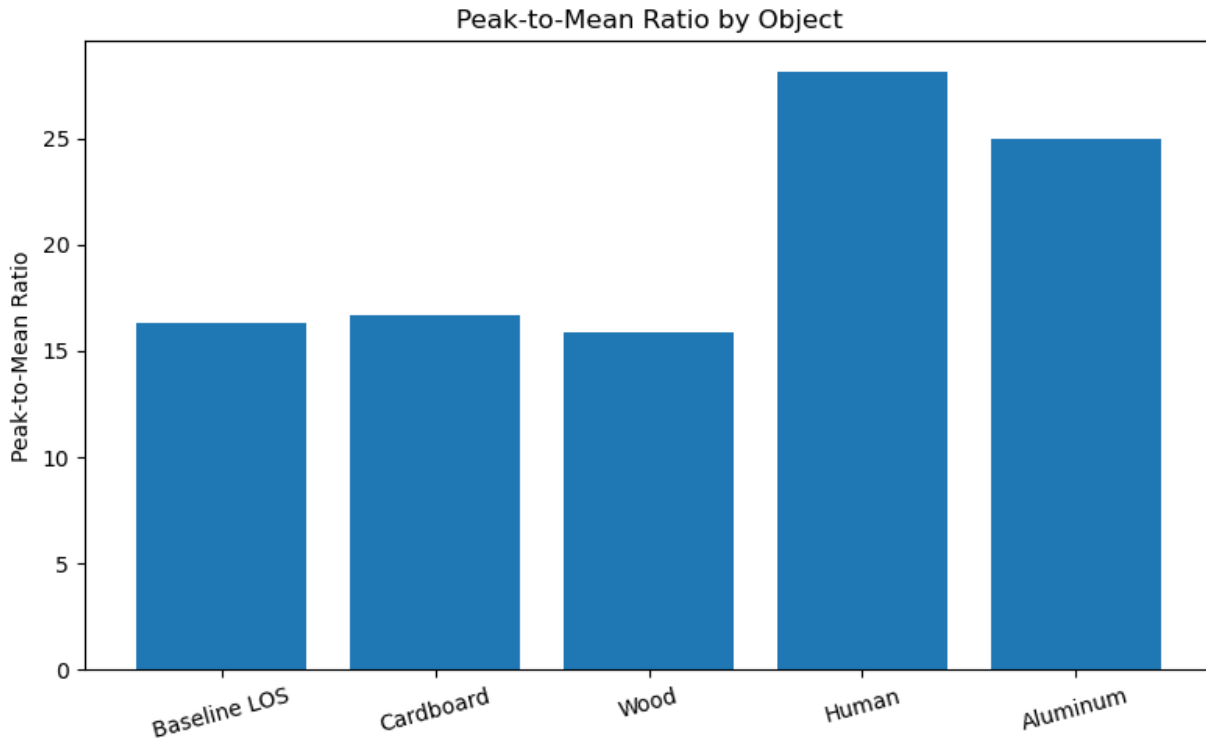


Figure 8. Peak-to-mean ratio comparison across obstruction materials. Metallic and human-body obstructions measurably altered dominant propagation behavior and multipath response.

These results further illustrate how obstruction material properties can significantly influence reflection behavior, attenuation, and multipath structure within indoor RF environments.

Interestingly, the human-body obstruction scenario produced the highest observed peak-to-mean ratio at approximately 28, exceeding even the aluminum obstruction scenario at approximately 25. This result was somewhat unexpected because the human body behaves primarily as a lossy dielectric medium rather than a strong metallic reflector. One possible explanation is that the human-body obstruction altered the surrounding multipath geometry in a manner that constructively redirected or concentrated reflected energy toward the receiver. The high water content of the human body may also have contributed to complex absorption and scattering behavior within the indoor propagation environment. Additional controlled experiments would be required to fully characterize this effect.

6.5 Bandwidth-Dependent Resolution Analysis

Bandwidth comparison testing was performed at:

- 250 kHz
- 1 MHz

- 5 MHz

Results demonstrated improved delay resolution at higher bandwidths.

Higher bandwidth configurations produced sharper correlation peaks, improved temporal resolution, and better multipath separation capability. Measured CIR full-width half-maximum (FWHM) decreased from approximately 8 μs at 250 kHz to 2 μs at 1 MHz and approximately 0.6 μs at 5 MHz, quantitatively demonstrating the improvement in temporal resolution achieved through wider bandwidth operation. Theoretical delay resolution scales approximately as $\Delta\tau \approx 1/B$, corresponding to approximately 4 μs at 250 kHz, 1 μs at 1 MHz, and 0.2 μs at 5 MHz. The measured FWHM results followed the same overall trend while exhibiting broader effective peak widths due to practical SDR filtering, finite observation windows, and smoothing effects.

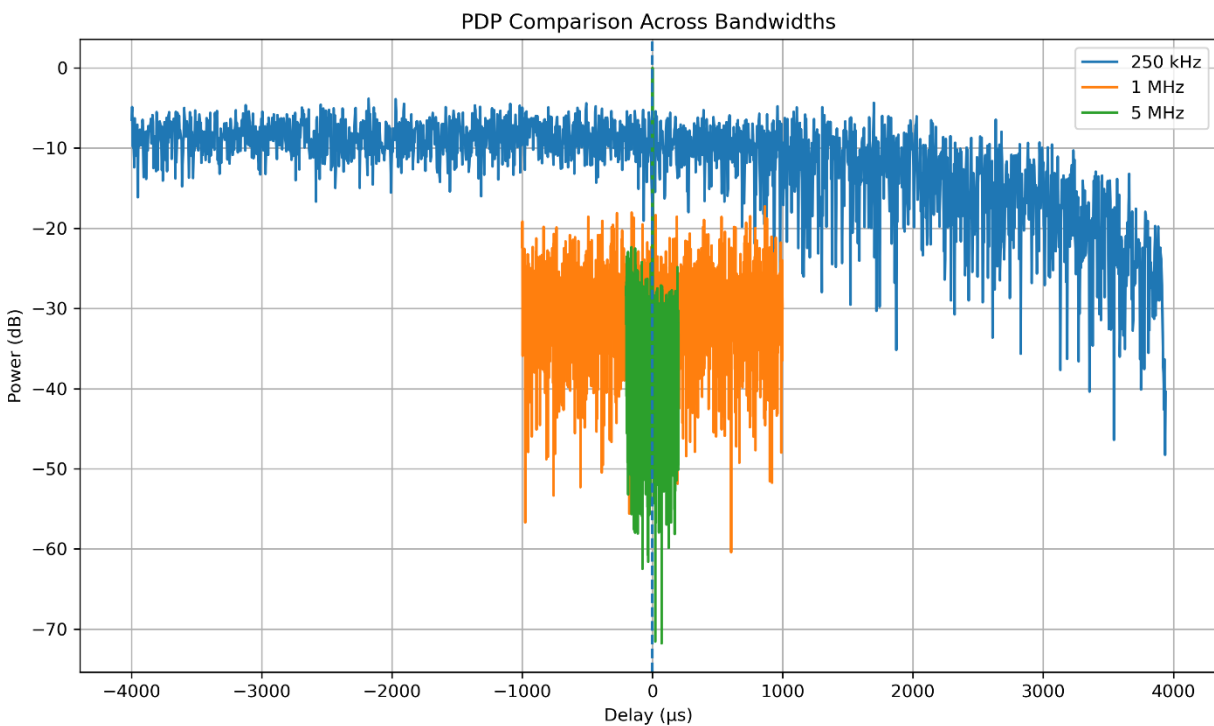


Figure 9. PDP overlays comparing 250 kHz, 1 MHz, and 5 MHz bandwidth configurations. Higher bandwidth operation produced sharper correlation peaks and improved multipath separation capability.

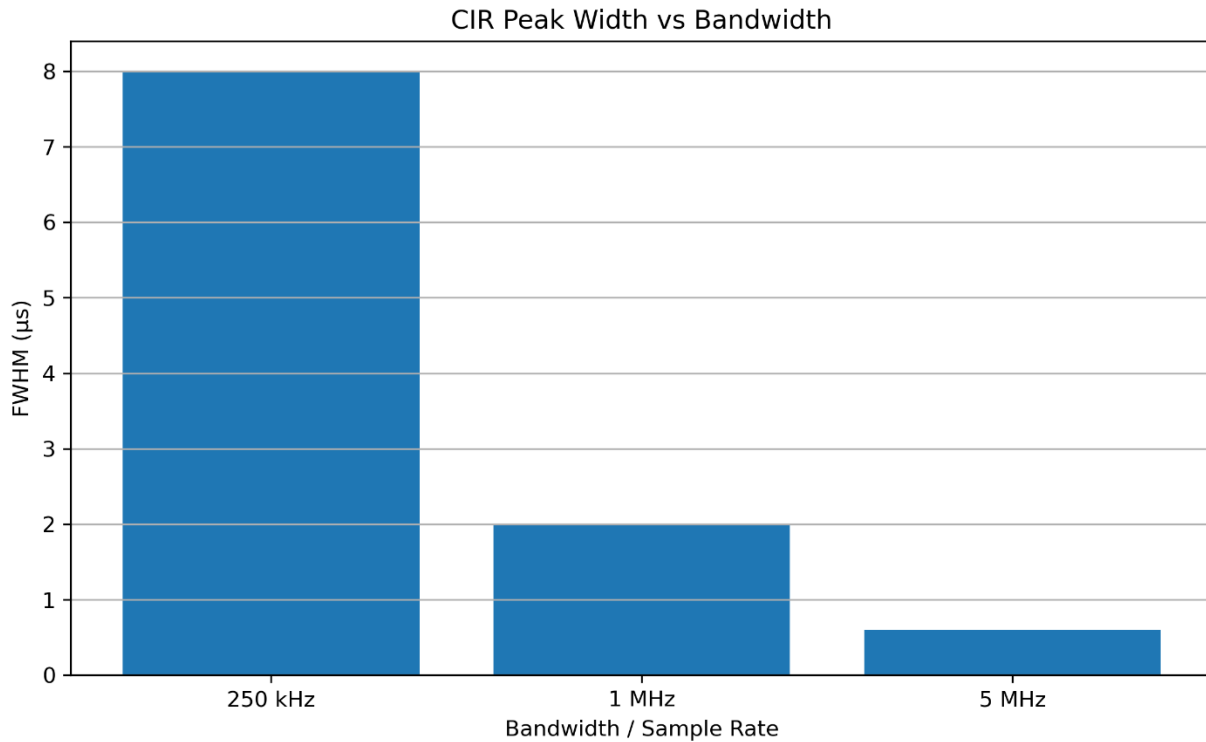


Figure 10. CIR peak width comparison across bandwidth configurations. Increasing bandwidth improved temporal resolution and reduced effective impulse-response width.

The narrowing of the effective CIR peak width at higher bandwidths further confirms the expected inverse relationship between bandwidth and temporal resolution.

These results align with theoretical expectations that wider bandwidth improves delay-domain resolution. [8], [9]

6.6 Frequency-Selective Analysis

The FFT-based channelizer successfully demonstrated sub-band response analysis. The FFT-based channelizer produced frequency-response overlays, sub-band gain visualizations, and delay-versus-frequency heatmaps that revealed measurable frequency-selective fading behavior across the occupied bandwidth.

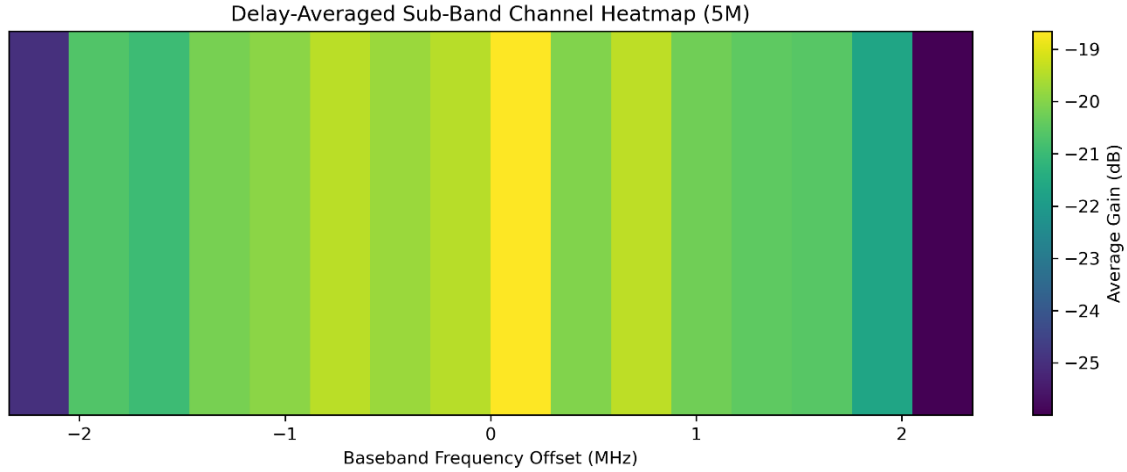


Figure 11. Delay-versus-frequency heatmap generated from FFT-based channelization. Variations across sub-bands indicate measurable frequency-selective fading behavior within the indoor propagation environment.

The heatmap visualization demonstrates that channel response varied across frequency, confirming measurable frequency-selective fading within the indoor propagation environment. Each heatmap bin represents the average normalized sub-band response magnitude across the analyzed delay window.

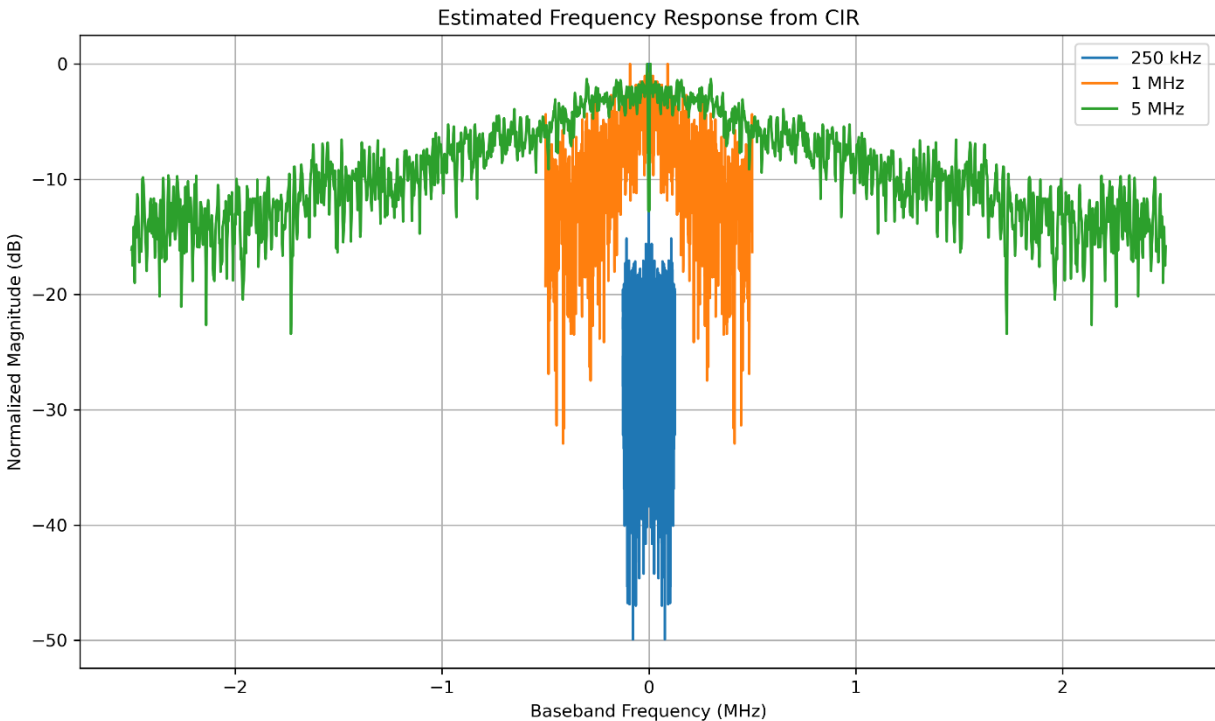


Figure 12. Estimated channel frequency response across multiple bandwidth configurations. Frequency-selective fading behavior was observable across several sub-bands.

Frequency-selective fading behavior was observable across multiple sub-bands, validating the ability of the FFT-based channelizer to capture variations in channel response across frequency. [8], [10]

Light moving-average smoothing was applied to selected comparative overlays to improve visualization readability while preserving underlying propagation structure and relative trend behavior.

6.7 Coherence Bandwidth Estimation

Coherence bandwidth analysis provided an estimate of the frequency range over which the channel response remained correlated. Coherence bandwidth estimates were derived from the observed delay spread behavior across multiple propagation scenarios. Environments with broader multipath dispersion generally exhibited reduced frequency correlation, indicating increased frequency-selective fading behavior. The baseline LOS scenario produced the largest estimated coherence bandwidth at approximately 2.04 MHz, while the 5.25 ft LOS scenario produced the lowest estimate at approximately 1.67 MHz, indicating increased frequency-selective fading behavior under stronger multipath dispersion conditions. These observations align with the expected inverse relationship between delay spread and coherence bandwidth. [1], [9]

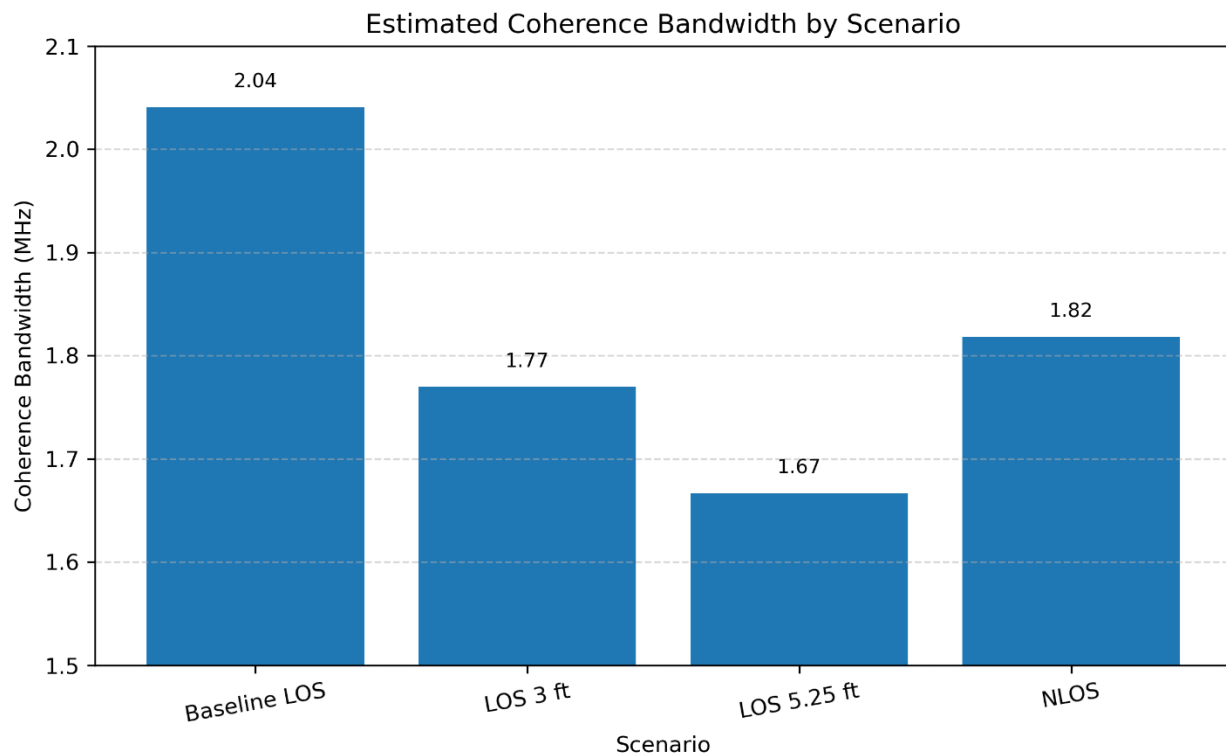


Figure 13. Estimated coherence bandwidth across representative propagation scenarios. Environments exhibiting increased multipath dispersion generally produced lower coherence bandwidth values, indicating stronger frequency-selective fading behavior.

The results support the relationship between delay spread and channel frequency selectivity. Future work could additionally estimate coherence bandwidth directly from the measured frequency-response correlation behavior observed in the FFT-based channelizer outputs to compare frequency-domain and delay-domain estimation approaches.

Interestingly, the measured NLOS coherence bandwidth remained slightly higher than the indoor LOS 3 ft and 5.25 ft scenarios. This behavior likely resulted from the specific geometry of the tested NLOS environment, where propagation may have been dominated by one or more relatively strong reflected paths rather than highly diffuse multipath dispersion.

6.8 Key Experimental Findings

The completed experimental campaign revealed several consistent propagation behaviors across the tested environments.

Indoor environments consistently produced richer multipath structure and higher Correlation-Domain RMS Spread than simpler LOS conditions. Object obstructions measurably altered CIR structure, peak-to-mean ratio, correlation floor behavior, and overall reflection characteristics across the tested scenarios. Correlation floor metrics were computed as the average normalized correlation magnitude outside the dominant CIR peak region and used as a relative indicator of background multipath and noise behavior. Higher bandwidth configurations improved temporal resolution, multipath separation capability, and correlation peak sharpness, while relatively small receiver position changes significantly altered dominant-path behavior, delay spread, and overall channel metrics. Frequency-selective fading behavior was also observable across multiple sub-bands, validating the effectiveness of the implemented FFT-based channelization approach. These findings demonstrate the highly environment-dependent and spatially sensitive nature of indoor wireless propagation.

Representative propagation metrics extracted from several experimental scenarios are summarized in Table 2.

Scenario	Relative Correlation Delay (μs)	Correlation-Domain RMS Spread (μs)	Peak-to-Mean Ratio	Mean Floor
Indoor LOS – 6 ft	N/A	114.32	6.05	0.165
Indoor LOS – 10 ft	N/A	105.18	4.20	0.238
Indoor LOS – 15 ft	-23.14	118.81	7.55	0.133
Outdoor LOS	13.98	100.50	4.17	0.240
Outdoor Obstructed	-12.82	109.47	6.27	0.160
Baseline LOS Object Test	N/A	111.35	16.35	0.061
Cardboard Obstruction	N/A	115.24	16.71	0.060
Wood Obstruction	N/A	110.86	15.85	0.063
Human Obstruction	N/A	114.03	28.16	0.036
Aluminum Obstruction	N/A	109.80	24.93	0.040

Table 2. Comparative propagation metrics extracted across representative indoor, outdoor, distance-variation, and obstruction-testing scenarios. Relative correlation delay values represent SDR correlation-domain alignment offsets intended for comparative analysis rather than calibrated physical propagation delay measurements.

Relative correlation delay values were extracted only for comparative scenarios where dominant correlation-peak offset variation was explicitly analyzed. Several indoor scenarios used aligned or normalized reference timing and, therefore, did not produce meaningful relative delay-offset measurements.

These comparative metrics reinforce the observed relationship between environmental obstruction, delay spread behavior, and dominant-path variability across the tested propagation scenarios. These values represent relative correlation-domain dispersion metrics derived from SDR sample-index alignment and processing-window behavior rather than calibrated absolute physical multipath delay measurements.

7. Discussion

7.1 Project Success Evaluation

The implemented SDR channel sounding platform successfully achieved all primary project objectives, including reliable PN-sequence transmission, repeatable IQ capture, cross-correlation-based CIR estimation, and PDP visualization across multiple propagation environments.

Additionally, several stretch objectives were implemented, including:

- FFT-based channelization
- Bandwidth comparison analysis
- Frequency-selective visualization
- Coherence bandwidth estimation

7.2 Technical Challenges

Several implementation challenges were encountered during development:

- **SDR Synchronization:** Timing alignment between independent SDR devices required careful correlation-based processing.
- **Environmental Variability:** Indoor propagation conditions introduced variability between captures due to reflections and environmental changes.
- **Noise and Stability:** Signal averaging and filtering were required to improve CIR clarity and reduce noise sensitivity.

These implementation challenges are representative of practical SDR-based wireless experimentation and influenced several architectural decisions throughout the project.

7.3 Engineering Tradeoffs

Several architectural decisions prioritized implementation reliability and experimental repeatability over maximum system complexity. In particular, the choice to use offline processing significantly simplified synchronization debugging, enabled repeatable analysis workflows, and reduced real-time computational constraints during development.

The project required balancing implementation complexity, computational cost, schedule constraints, and measurement quality throughout system development.

The decision to prioritize a modular offline-processing workflow significantly improved development speed and debugging efficiency.

7.4 Lessons Learned

Several practical lessons emerged throughout the implementation and experimental phases of the project. One important observation was that synchronization and timing stability represented a larger challenge than initially anticipated when operating independent SDR platforms without external reference locking. These challenges significantly influenced the decision to prioritize offline processing and correlation-based alignment methods throughout system development.

The project also demonstrated that indoor RF propagation behavior can vary substantially over relatively small spatial distances. In several experiments, minor receiver repositioning produced measurable changes in delay spread, dominant path strength, and correlation behavior, reinforcing the highly geometry-dependent nature of indoor wireless channels.

Additionally, the implementation process highlighted the importance of modular signal-processing workflows, repeatable experimental procedures, and visualization techniques for interpreting complex propagation behavior. The use of Python-based offline processing significantly improved debugging flexibility and accelerated experimental iteration throughout system development.

The project also demonstrated the importance of balancing theoretical signal-processing concepts with practical implementation constraints. While many wireless propagation concepts can be modeled analytically, real-world SDR experimentation requires iterative debugging, parameter tuning, visualization refinement, and experimental validation to obtain stable and interpretable results.

7.5 Comparison to Expected Theory

Experimental observations were broadly consistent with established indoor multipath propagation theory and expected SDR-based channel sounding behavior. NLOS environments produced increased delay spread, higher bandwidth improved temporal resolution, obstructions altered multipath structure, and transmitter-receiver separation affected dominant path strength.

The observed reduction in CIR peak width with increasing bandwidth aligned closely with the theoretical delay-resolution relationship $\Delta\tau \approx 1/B$ introduced in Section 2.4. However, the measured full-width half-maximum (FWHM) values remained broader than the ideal theoretical limits. For example, the measured CIR peak width at 1 MHz was approximately 2 μs compared to the idealized theoretical resolution of approximately 1 μs , while the 250 kHz and 5 MHz measurements also exhibited broader effective widths than predicted. These deviations likely resulted from practical SDR filtering effects, finite observation windows, correlation-window behavior, smoothing operations, and residual synchronization uncertainty between independent SDR platforms.

Similarly, environments exhibiting increased Correlation-Domain RMS Spread generally produced reduced coherence bandwidth estimates, consistent with the expected inverse relationship between delay spread and channel frequency selectivity. The FFT-based channelizer results further supported this relationship by revealing measurable frequency-selective fading behavior across multiple sub-bands.

The observed non-monotonic spatial variation in dominant-path strength and Correlation-Domain RMS Spread also aligned with expected indoor multipath interference behavior, where relatively small receiver position changes can alter constructive and destructive interference relationships between reflected propagation paths.

Overall, the measured propagation behavior aligned closely with expected theoretical indoor multipath characteristics and validated the effectiveness of the implemented SDR-based sounding architecture.

7.6 System Limitations

Several limitations should be considered when interpreting the experimental results.

Independent SDR Timing

The transmitter and receiver operated as independent SDR platforms without external phase locking or precision timing synchronization. Correlation-based alignment techniques mitigated timing offsets; however, some residual timing uncertainty remained. Residual timing offsets following correlation alignment were typically within approximately ± 2 –5 samples, depending on capture conditions. For example, at a 1 MHz sample rate, a residual timing uncertainty of ± 2 samples corresponds to approximately ± 2 μ s of relative timing uncertainty.

Environmental Variability

Indoor RF environments are inherently dynamic and highly geometry dependent. Small environmental changes, object movement, and receiver positioning variations could measurably influence channel behavior and multipath structure.

Limited Measurement Scale

The project focused primarily on controlled indoor propagation scenarios and limited outdoor comparisons. Larger-scale outdoor propagation analysis was intentionally deferred to maintain project feasibility within the available timeline.

Offline Processing Architecture

The implemented system relied on offline processing rather than real-time streaming analysis. While this significantly improved debugging flexibility and experimental repeatability, it limited real-time channel observation capability.

Antenna Placement Repeatability

Antenna orientation and small positional variations between captures may also have introduced minor measurement variability throughout the experimental campaign.

7.7 Practical Relevance

The experimental propagation behaviors observed throughout this project are directly relevant to modern wireless communication systems operating in complex RF environments. Multipath propagation, delay spread, and frequency-selective fading significantly influence the performance of Wi-Fi networks, tactical communication systems, radar platforms, autonomous sensing systems, and emerging adaptive wireless architectures.

The results demonstrated that relatively small environmental and positional changes produced measurable variations in channel response, dominant path behavior, and

coherence bandwidth. These observations reinforce the importance of experimental channel characterization during wireless system design, deployment, and validation.

In addition to its propagation analysis capabilities, the implemented SDR-based architecture demonstrates how low-cost and commercially accessible SDR platforms can support meaningful RF experimentation without requiring specialized proprietary channel sounding equipment. The modular design approach also provides educational and research value by enabling flexible experimentation with signal processing algorithms, synchronization techniques, waveform design, and future wireless system extensions.

The developed platform, therefore, serves not only as a propagation measurement tool but also as a flexible experimental framework for future SDR, RF, and wireless communications research.

8. Future Work

8.1 MIMO Channel Sounding

Expanding the system to a multi-antenna architecture would enable:

- Spatial diversity analysis
- Beamforming experimentation
- MIMO propagation characterization

This represents the most valuable future extension because it would substantially increase the system's applicability to modern wireless communications research.

Future MIMO implementation could leverage phase-coherent SDR synchronization using shared external clock references or GPS-disciplined oscillators to support spatial channel estimation and beamforming analysis using multi-channel SDR configurations.

8.2 Doppler Analysis

Introducing controlled receiver motion would support:

- Doppler shift estimation
- Time-varying channel analysis
- Mobility modeling

This would enable the evaluation of dynamic propagation environments and moving wireless platforms. Delay and Doppler spread behavior are commonly used to characterize dynamic wireless channels and mobile propagation environments [5]. Controlled receiver motion using linear translation or rotating reflector experiments could additionally support repeatable Doppler spread characterization under dynamic indoor propagation conditions.

8.3 Real-Time Processing

Future work could transition portions of the processing chain to real-time operation using:

- GNU Radio integration
- Streaming visualization
- FPGA-assisted acceleration

Real-time capability would significantly improve the system's usability for live propagation analysis and adaptive wireless experimentation.

Modern USRP-based channel sounders increasingly incorporate onboard FPGA processing and real-time averaging techniques to reduce data throughput requirements and improve measurement quality.

9. Conclusion

This project successfully implemented an SDR-based wireless channel sounder for indoor multipath characterization. Using two USRP B200 SDR devices operating at 2.45 GHz, the system successfully transmitted a PN sequence, captured IQ data, estimated channel impulse responses, and generated meaningful propagation visualizations.

The completed system validated key wireless propagation concepts, including multipath delay spread, bandwidth-dependent resolution, obstruction effects, and frequency-selective fading behavior. Experimental results revealed measurable differences between LOS and NLOS environments and confirmed the effectiveness of SDR platforms for propagation experimentation.

Beyond meeting the original MVP requirements, the project successfully incorporated advanced analysis techniques, including FFT-based channelization and coherence bandwidth estimation. The resulting platform provides a modular foundation for future RF propagation and wireless systems research.

Additional implementation details, supplemental figures, repository structure, and project-specification alignment documentation are provided in the appendices to support reproducibility and future system extension.

The observed variation in delay spread, coherence bandwidth, and dominant-path behavior across relatively small environmental and spatial changes reinforces the importance of adaptive channel-aware wireless system design. These results demonstrate how even low-cost SDR-based measurement platforms can provide meaningful insight into real-world propagation behavior relevant to modern Wi-Fi, tactical communications, OFDM-based wireless systems, and future adaptive RF architectures. The implemented system also demonstrates how accessible SDR platforms can bridge theoretical wireless communications concepts with practical experimental propagation analysis.

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Appendix A – Repository Structure

```
chelle@Chelle:~/SDR-Channel-Sounder-Capstone$ tree -L 2
.
├── README.md
├── presentation
│   ├── mvp_demo_presentation.pdf
│   └── project_overview_brief.pdf
├── requirements.txt
├── results
│   ├── analysis
│   ├── bandwidth_tests
│   ├── coherence_v7
│   ├── distance_tests
│   ├── object_tests
│   └── outdoor_tests
├── scripts
│   ├── analysis
│   ├── sdr_capture
│   └── signal_processing
└── 13 directories, 4 files
```

The repository is organized into acquisition, signal-processing, analysis, and results directories to support experiment reproducibility and modular workflow execution.

Folder	Purpose
scripts/sdr_capture	SDR transmission/acquisition
scripts/signal_processing	CIR/PDP extraction
scripts/analysis	Comparative propagation analysis
results/	Experimental outputs

Appendix B – Experimental Result Figures

This appendix contains supplemental experimental figures and secondary propagation metrics that support the primary analysis presented in Section 6. These supplemental figures provide additional insight into secondary propagation metrics and visualization outputs that were not included in the primary results discussion.

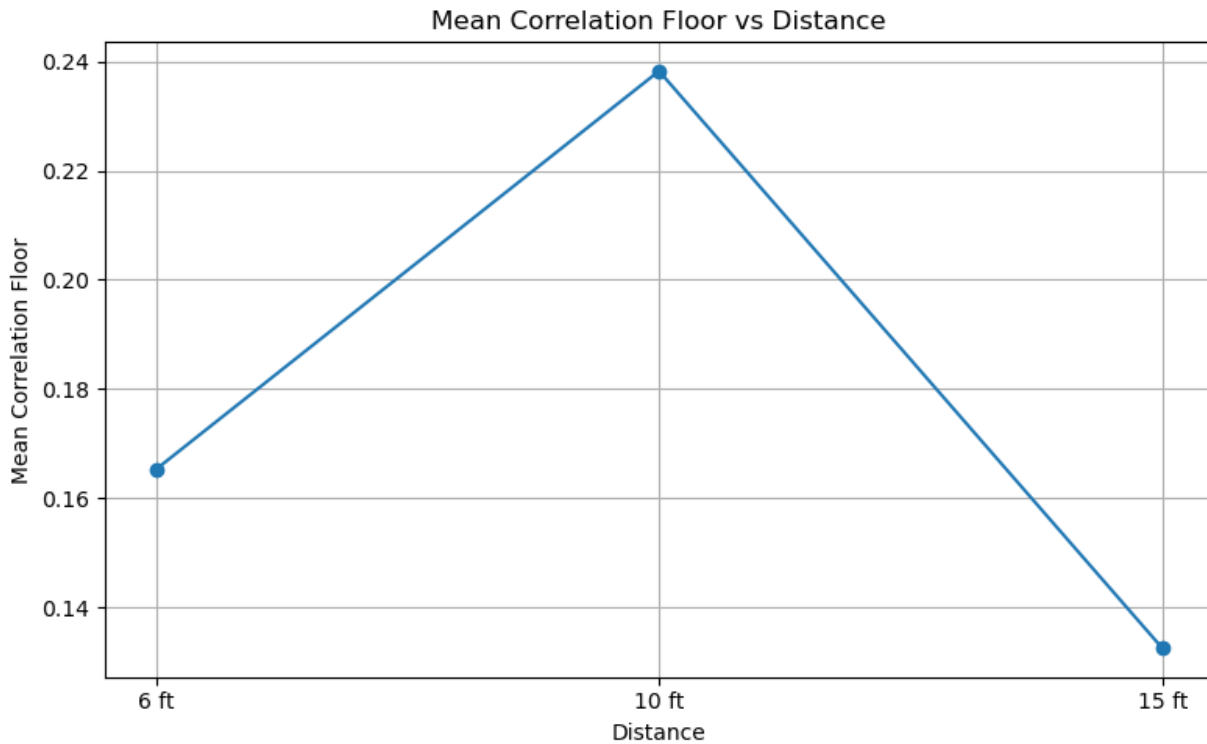


Figure B1. Mean correlation floor variation across indoor distance measurements. Environmental geometry and multipath structure contributed to non-uniform background correlation behavior.

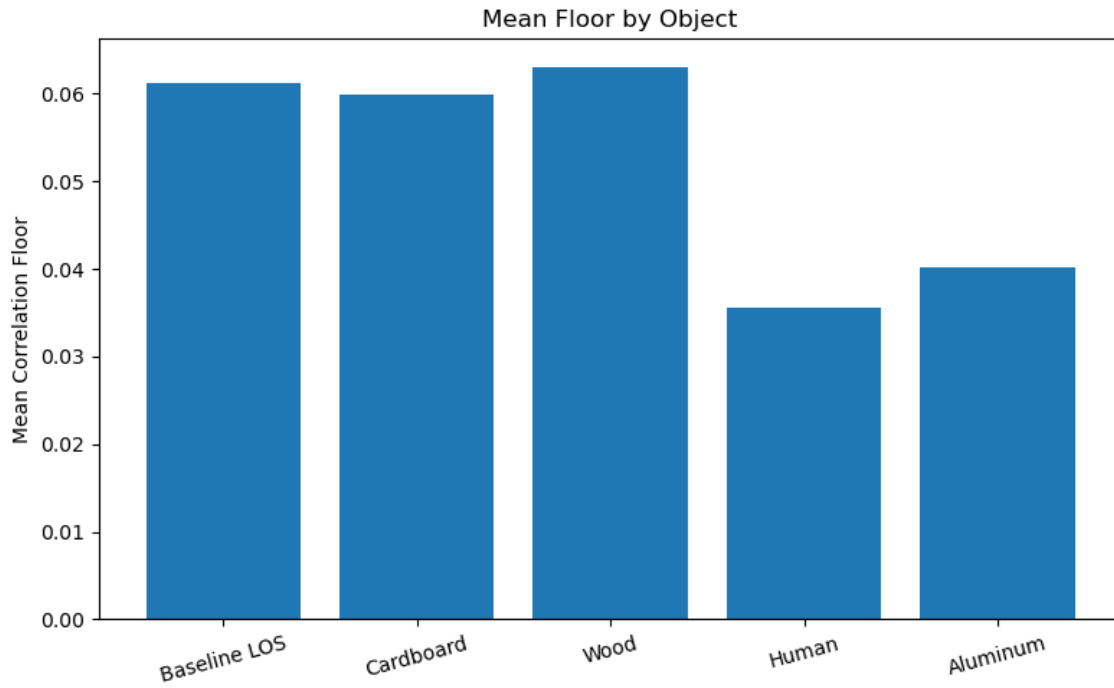


Figure B2. Mean correlation floor comparison across obstruction materials. Human-body and metallic obstructions produced measurable changes in background correlation structure.

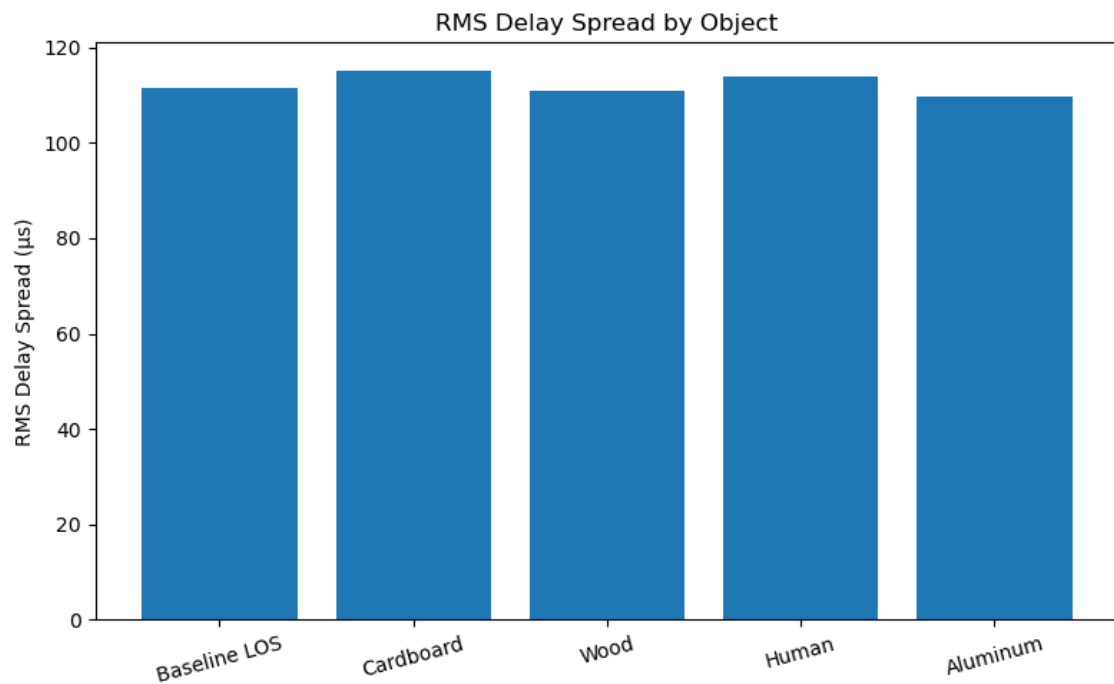


Figure B3. Correlation-Domain RMS Spread comparison across obstruction materials. Environmental obstructions introduced measurable variations in multipath dispersion behavior.

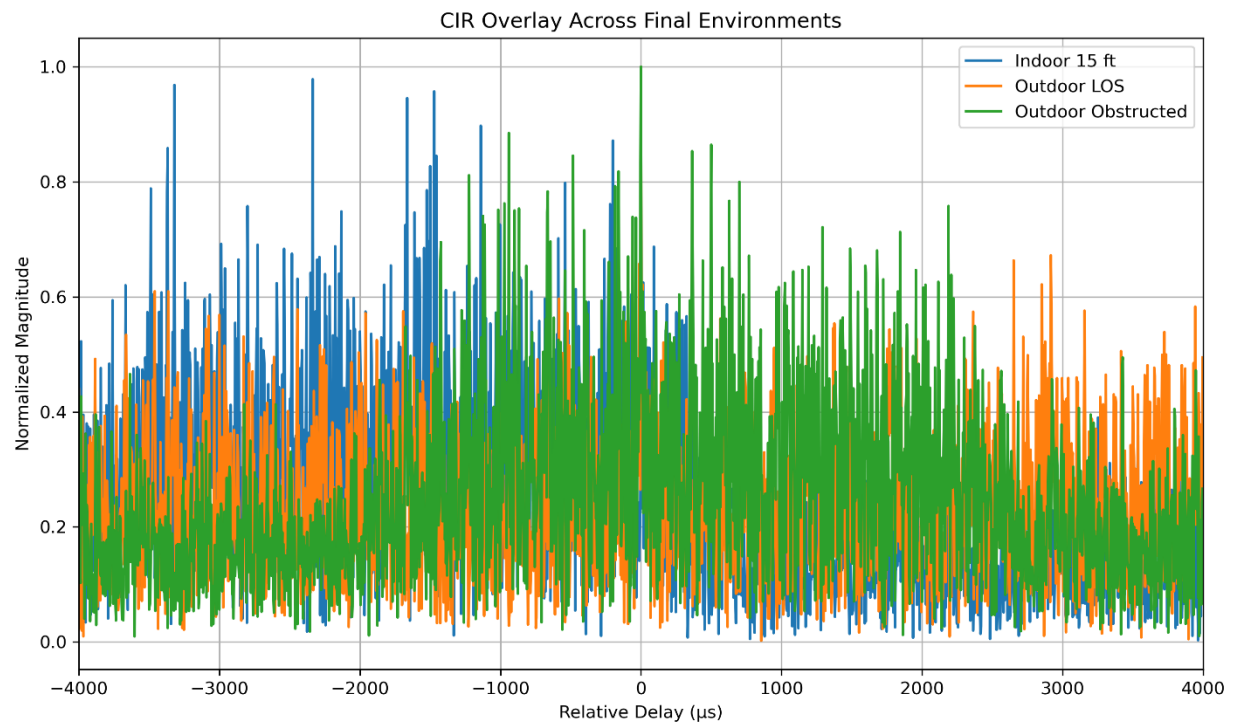


Figure B4. Raw unsmoothed CIR overlays for multiple propagation environments prior to visualization smoothing.

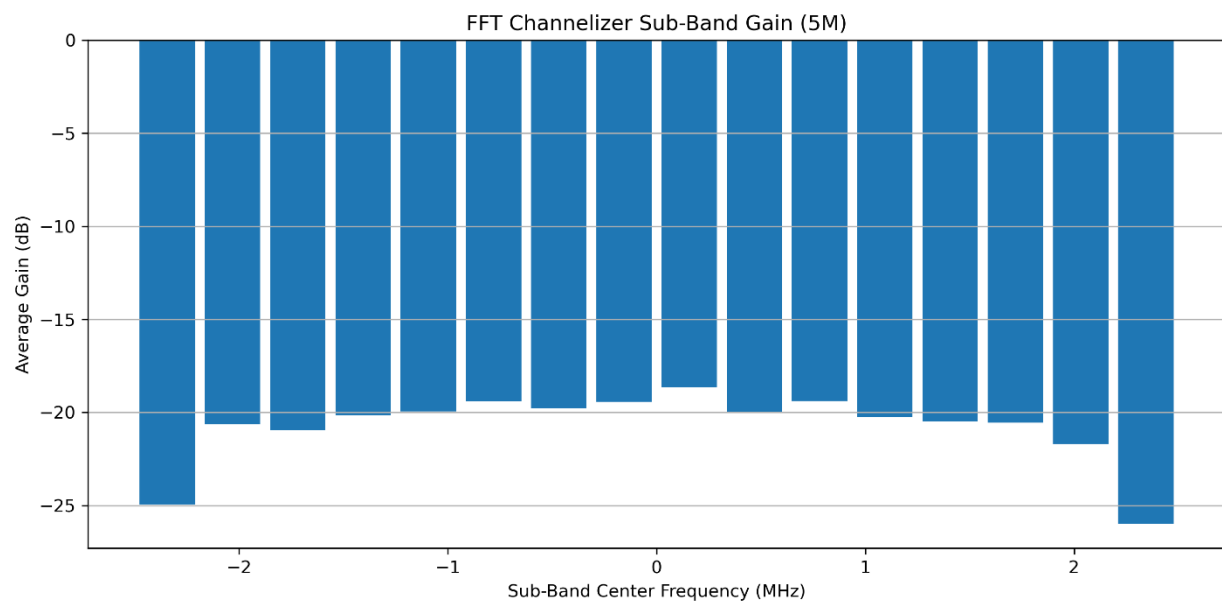


Figure B5. Supplemental FFT-based sub-band channel response comparison illustrating frequency-selective propagation behavior across multiple channel segments.

Appendix C – Project Specification Alignment

This project successfully aligned with the approved project specification, including:

- SDR-based architecture
- Indoor propagation analysis
- PN-sequence channel sounding
- CIR and PDP generation
- LOS/NLOS comparison
- Signal processing enhancements
- Stretch objective implementation

The final implementation exceeded the original MVP scope through the inclusion of FFT-based frequency-selective analysis and advanced comparative visualizations.

Requirement	Status	Evidence
OTA PN sounding	Complete	Section 6.1
CIR extraction	Complete	Figures 3–4
Environmental testing	Complete	Sections 6.2–6.4
Bandwidth comparison	Complete	Section 6.5
Frequency-selective analysis	Complete	Section 6.6

Table C1. Project specification completion summary.

Metric	Purpose
CIR	Multipath timing
PDP	Delay-domain power
Correlation-Domain RMS Spread	Channel dispersion
Peak-to-Mean Ratio	Dominant path strength

Table C2. Primary channel metrics used throughout the analysis.